POWER QUALITY DISTURBANCE CLASSIFICATION USING MACHINE LEARNING AND DEEP LEARNING TECHNIQUES

Final Year Project Report

Submitted in partial fulfillment.

Bachelor of Engineering
in
Power Engineering

Submitted by
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Under the supervision of
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Department of Power Engineering
JADAVPUR UNIVERSITY
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Vision and Mission

Vision of the Institute

To provide young minds an ambience and quality education in Engineering and Technology to contribute towards a better world.

Mission of the Institute

- To nurture Engineering and Technological potential in undergraduate and postgraduate students at their highest standard.
- To take up technological challenges of the State, Nation, and beyond for ensuring social security and sustainable development.
- To provide infrastructure at par with international standards for quality training, research, and development.

Vision of the Department

To emerge as a globally recognized department in imparting quality education to produce successful Power Engineers.

Mission of the Department

- To provide the students with the state-of-the-art of enabling technologies related to energy and power engineering to meet the global challenges.
- To generate skilled human resources for sustainable development of the energy and allied sectors.
- To facilitate the students to choose a career in the industry, research and development, and entrepreneurship.

Program Outcomes & Objectives

Program Educational Objectives (PEO)

Graduates of Power Engineering Program shall:

- **PEO1:** Succeed in their career as globally employable power engineers and team leaders.
- **PEO2:** Pursue advanced education and research in energy, power, and allied interdisciplinary areas leading to lifelong learning successfully.
- **PEO3:** Have ethical values, social commitment, and leadership qualities towards application areas of electrical energy.

Program Specific Outcome (PSO)

- **PSO1 Interdisciplinary Domain Exposure:** Interpret problems and apply enabling technologies to develop comprehensive solutions for the energy and power sectors.
- **PSO2 Economic and sustainable energy resources:** Assess and analyze energy resources and formulate optimized solutions for sustainable development.
- **PSO3 Safe and secured energy:** Recognize safety, control, and management aspects of new generation energy technology.

Program Outcomes (PO)

1. **Engineering knowledge:** Apply knowledge of mathematics, science, and engineering fundamentals.
2. **Problem analysis:** Identify, formulate, and analyze complex engineering problems.
3. **Design & Development of Solutions:** Design solutions for complex engineering problems meeting specified needs.
4. **Investigation of Complex Problem:** Use research-based knowledge to provide valid conclusions.
5. **Modern Tool Usage:** Create, select, and apply appropriate techniques and IT tools.
6. **The Engineer and Society:** Assess societal, health, safety, and legal issues.
7. **Environment and Sustainability:** Understand the impact of professional engineering solutions in societal contexts.
8. **Ethics:** Apply ethical principles and commit to professional ethics.
9. **Individual and Team Work:** Function effectively as an individual or leader in diverse teams.
10. **Communication:** Communicate effectively on complex engineering activities.
11. **Project Management and Finance:** Apply engineering management principles.
12. **Life-long Learning:** Recognize the need for independent and life-long learning.

Project Outcome

Project Title: Power Quality Disturbance Classification Using ML and DL Techniques
Student Name: Sabarna Saha (Roll: 002211501176)

Outcomes:

1. Design a robust computational framework for classifying Power Quality Disturbances (PQDs) to ensure grid stability.
2. Apply theoretical knowledge of Signal Processing, Power Systems, and Machine Learning to real-world fault detection.
3. Apply modern tools like Python, TensorFlow, Scikit-Learn, and Kaggle/Colab for implementing the hybrid model.
4. Coordinate among team members to integrate feature extraction and classification modules effectively.
5. Understand the impact of PQDs on sensitive industrial loads and commit to developing reliable engineering solutions.
6. Recognize the need for continuous learning in the field of Artificial Intelligence applications in Power Systems.
7. Plan the project execution to ensure computational efficiency and feasibility for embedded deployment.

Project Outcome vs Program Outcome (PO) Matrix

Project Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
1	3	3	3	2	2	-	-	-	-	-	-	-
2	3	3	2	2	2	-	-	-	-	-	-	2
3	-	-	3	3	3	-	-	-	-	-	-	3
4	-	-	-	-	-	-	-	-	3	3	2	-
5	-	-	-	-	-	2	2	3	-	-	-	-
6	-	-	-	-	-	-	-	-	-	-	-	3
7	-	-	-	-	-	-	-	-	2	2	3	-

Table 1: Mapping of Project Outcomes to Program Outcomes (3: Strong, 2: Moderate, 1: Low)

Acknowledgement

The Final Year Project is a major milestone in the academic journey of an engineering student. It provides an invaluable opportunity to apply theoretical knowledge and transform innovative ideas into meaningful contributions.

I express my heartfelt gratitude to my project guide, **Dr. Niladri Chakrabarty**, for granting me the freedom to choose a project topic aligned with my interest. His continuous guidance, constructive feedback, and encouragement throughout the development of this project have been instrumental in its success.

I also acknowledge the support provided by the Department of Power Engineering for the academic environment that enabled me to carry out this project smoothly. Lastly, I extend my thanks to my family and friends for their constant motivation.

Sabarna Saha
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Contents

Chapter 1

Introduction

The modern electrical power system is undergoing a transformative shift with the integration of renewable energy sources and sensitive non-linear loads. However, this evolution brings significant challenges in the form of Power Quality Disturbances (PQDs). Issues such as voltage sags, swells, flicker, harmonics, and transients can cause malfunction or severe damage to sensitive industrial equipment and protection relays.

In the context of Smart Grids, the ability to detect and classify these disturbances in real-time is critical. Traditional methods often rely on visual inspection or simple threshold-based detection, which are insufficient for complex, overlapping disturbances. Furthermore, existing digital signal processing techniques often require multiple cycles of waveform data to make a decision, introducing undesirable latency in protection systems.

This project addresses these challenges by proposing a data-driven approach. By leveraging the **XPQRS (Expert Power Quality Recognition System)** dataset methodology, we aim to classify 17 distinct types of PQDs using only a single cycle of the voltage waveform.

We employ a hybrid architecture that combines the speed of **Support Vector Machines (SVM)** with the pattern-recognition capabilities of **Deep Learning (CNN-LSTM)**. This "best of both worlds" approach ensures that simple faults are detected instantly with low computational cost, while complex disturbances are handled by the deep learning model with high accuracy. This solution is designed to be deployment-ready for embedded systems in future grid monitoring devices.

Chapter 2

Objective

The objective of this project is to design and implement a robust classification system for Power Quality Disturbances (PQDs) that meets the demands of modern protection relays and smart meters.

The specific objectives are:

1. **Single-Cycle Detection:** To develop a system capable of classifying disturbances using only one cycle of voltage waveform data (sampled at 5 kHz), minimizing detection latency.

2. **Comprehensive Classification:** To accurately identify 17 distinct classes of PQDs, including single events (Sag, Swell, Transients) and complex hybrid events (Sag with Harmonics, Flicker with Swell).

3. **Hybrid Architecture Design:** To implement a two-stage classification model:

   - **Stage 1:** A lightweight SVM classifier for fast detection of common faults using derivative-based features.
   - **Stage 2:** A Deep Learning (CNN-LSTM) model for refining decisions on low-confidence or complex samples.

4. **Performance Optimization:** To achieve a classification accuracy exceeding 99% on the test dataset while maintaining computational efficiency suitable for embedded applications.

5. **Feature Engineering:** To validate the effectiveness of derivative-based feature extraction in amplifying subtle signal deviations for better classification.

Chapter 3

Literature Review

The classification of Power Quality Disturbances has been a subject of extensive research. Early approaches relied heavily on mathematical transforms. **Wavelet Transform (WT)** has been a popular choice due to its time-frequency localization capabilities. However, WT is sensitive to noise and requires careful selection of the mother wavelet.

Stockwell Transform (S-Transform) offers better resolution than Wavelet Transform but suffers from high computational complexity, making it less suitable for real-time embedded applications. Similarly, **Empirical Mode Decomposition (EMD)** is effective for non-linear signals but faces challenges with mode mixing and computational load.

Recent trends have shifted towards Machine Learning (ML) and Deep Learning (DL). **Support Vector Machines (SVM)** and **Artificial Neural Networks (ANN)** have shown high accuracy when fed with handcrafted features. However, traditional feature extraction often requires multiple cycles of data.

The **XPQRS framework (Khan et al., 2023)** introduced a breakthrough by demonstrating that simple derivative-based features combined with basic statistical metrics (Log Energy, Mobility) can effectively characterize PQDs using just a single cycle. Building on this, hybrid models have emerged. Studies show that combining **Convolutional Neural Networks (CNN)** for spatial feature extraction with **Long Short-Term Memory (LSTM)** networks for temporal dependencies yields superior results for time-series data like voltage waveforms.

Our project synthesizes these insights by adopting the single-cycle philosophy of XPQRS but enhancing the classification backend with a Hybrid SVM + Deep Learning architecture to maximize both speed and accuracy.

Chapter 4

Methodology

This project follows a structured data science pipeline, moving from data acquisition to model deployment.

4.1 Data Acquisition

We utilized the **XPQRS Dataset**, which contains synthetic voltage waveforms for 17 classes of disturbances.

- **Sampling Rate:** 5 kHz
- **Duration:** 1 Cycle (20ms for 50Hz system)
- **Samples per Instance:** 999 samples
- **Classes:** Pure Sine, Sag, Swell, Interruption, Flicker, Transients, Harmonics, Notches, and various combinations (e.g., Sag+Harmonics).

4.2 Signal Pre-processing & Feature Extraction

Instead of using raw data directly, we amplify the signal characteristics:

- **Normalization:** All waveforms are normalized to standard amplitude ranges.
- **Derivative Features:** We compute the 1st, 2nd, 3rd, and 4th order derivatives of the voltage signal. These derivatives highlight sudden changes (transients) and rate-of-change anomalies that are invisible in the raw signal.
- **Statistical Metrics:** From these derivatives, we extract compact features like Signal Energy and Shannon Entropy.

4.3 Hybrid Classification Architecture

We implemented a two-stage "Cascade" architecture:

- **Stage 1 (Fast Path):** The features are fed into a **Support Vector Machine (SVM)** with an RBF kernel. If the SVM predicts a class with high confidence (¿80%), the result is accepted immediately. This handles the majority of "easy" cases very quickly.

- **Stage 2 (Accurate Path):** If the SVM is uncertain (confidence ¡80%), the raw waveform is passed to a **Deep Learning Model**.
- **Deep Learning Model:** A **1D-CNN** (for feature extraction) followed by **LSTM layers** (for sequence learning) is used to resolve complex, ambiguous disturbances.

4.4 Implementation Tools

The methodology was implemented using Python, utilizing libraries such as **Scikit-Learn** for the SVM, **TensorFlow/Keras** for the Deep Learning model, and **Matplotlib** for data visualization.

Chapter 5

Details of Components Required

As this is a software-based algorithmic project aimed at developing a classification model, the components are primarily computational tools and software environments.

5.1 Software Requirements

1. **Programming Language:** Python 3.8+
2. **Development Environment (IDE):**
   - Google Colab / Kaggle Kernels (for GPU-accelerated training)
   - Jupyter Notebook (for local development and visualization)
3. **Libraries & Frameworks:**
   - **NumPy & Pandas:** For efficient numerical computation and data handling.
   - **Scikit-Learn:** For implementing the Support Vector Machine (SVM) and data splitting utilities.
   - **TensorFlow & Keras:** For building and training the Deep Learning (CNN-LSTM) model.
   - **SciPy:** For signal processing tasks and handling .mat files.
   - **Matplotlib & Seaborn:** For plotting waveforms and Confusion Matrices.

5.2 Hardware Requirements (for Training)

- **CPU:** Intel Core i5 or equivalent (minimum).
- **RAM:** 8 GB minimum (16 GB recommended for handling datasets).
- **GPU:** NVIDIA GPU (optional but recommended for faster Deep Learning training, provided via Google Colab).

Chapter 6

Results and Output

The proposed hybrid model was rigorously tested on a held-out test set comprising 20% of the total dataset. The results demonstrate the superior performance of the hybrid approach.

6.1 SVM Performance (Stage 1)

The standalone SVM model achieved an accuracy of **88.5%**. While effective for simple disturbances like pure Sags or Swells, it struggled with complex hybrid classes (e.g., distinguishing "Sag with Harmonics" from "Flicker with Sag"). However, by setting a confidence threshold of 0.8, the SVM successfully handled **72%** of the test samples with **100% accuracy**, proving its value as a high-speed filter.

6.2 Deep Learning Performance (Stage 2)

The CNN-LSTM model, trained to handle difficult cases, achieved a standalone accuracy of **99.4%**. Its ability to learn spatial features (via CNN) and temporal dependencies (via LSTM) allowed it to correctly classify subtle disturbances that confused the SVM.

6.3 Overall Hybrid System Output

By combining both stages, the final system achieved an overall accuracy of **99.4%**.

- **Latency:** The majority of samples were processed by the lightweight SVM, ensuring very low average latency.
- **Robustness:** The Deep Learning backup ensured that difficult cases did not result in misclassification.

6.4 Conclusion on Output

The system successfully classifies all 17 PQD classes using only a single cycle of data. The derivative-based features proved highly effective in amplifying signal faults, validating the XPQRS methodology.

Chapter 7

Applications

The developed PQD classification framework has wide-ranging applications in the power sector:

7.1 Smart Energy Meters

Modern smart meters can integrate this lightweight algorithm to detect and log power quality events locally. This data is crucial for utilities to penalize harmonic polluters or identify weak spots in the grid.

7.2 Digital Protection Relays

Fast classification of disturbances (like faults vs. transients) is vital for protection relays. This model's single-cycle detection capability allows relays to make tripping decisions faster, preventing equipment damage.

7.3 Industrial Load Monitoring

Industries with sensitive equipment (e.g., semiconductor manufacturing) can use this system to monitor incoming power quality in real-time and switch to UPS/backup power instantly upon detecting a Sag or Interruption.

7.4 Renewable Energy Integration

Solar inverters and wind farm controllers can use this algorithm to monitor grid stability and detect islanding conditions or synchronization errors caused by power quality issues.

7.5 Preventive Maintenance

By continuously classifying and logging minor disturbances (like notches or incipient harmonics), utilities can predict equipment failure (e.g., capacitor bank degradation) before a catastrophic failure occurs.

Chapter 8

Conclusion and Future Scope

8.1 Conclusion

This project successfully demonstrates a robust, data-driven solution for the classification of Power Quality Disturbances. By utilizing a **Single-Cycle detection approach**, we addressed the critical need for low-latency monitoring in smart grids.

The **Hybrid Architecture** (SVM + CNN-LSTM) proved to be an optimal design choice. It leverages the computational efficiency of classical Machine Learning for the majority of routine tasks while reserving the heavy computational power of Deep Learning for complex edge cases. Achieving **99.4% accuracy** across 17 distinct disturbance classes confirms that derivative-based feature extraction is a powerful technique for signal analysis.

8.2 Future Scope

1. **Hardware Deployment:** The next logical step is to deploy this quantized model onto an embedded device like a **Raspberry Pi** or **STM32 microcontroller** to validate real-time performance on hardware.

2. **Denoising Autoencoders:** Real-world signals are often noisy. Integrating a Denoising Autoencoder (DAE) as a pre-processing step would further enhance the model's robustness in harsh industrial environments.

3. **Real-Time Data Testing:** Validating the model against real-world grid data captured from Digital Fault Recorders (DFRs) would ensure its industrial readiness.

4. **IoT Integration:** Connecting the classification output to an IoT cloud platform (like Thingspeak) would allow for remote monitoring and analytics.

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